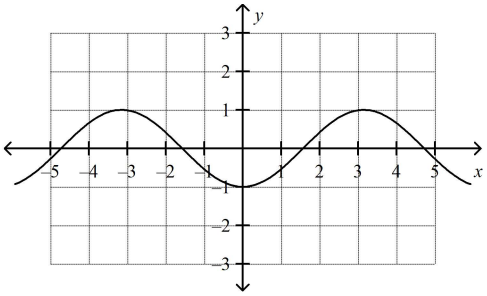
MULTIPLE CHOICE

1. Which function, where x is in radians, is represented by the graph shown below?



A. $y = -\cos x$
B. $y = \sin x$
C. $y = \cos x$
D. $y = -\sin x$

ANS: A DIF: Easy OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions KEY: graph | sinusoidal function

2. The range (in radians) of the graph of $y = \cos x$ is

A. $-1 \le y \le 1$
B. $0 \le y \le 2\pi$
C. $y \in R$
D. $-2\pi \le y \le 2\pi$

ANS: A DIF: Easy OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions KEY: range | sinusoidal function

3. The period (in degrees) of the graph of $y = \cos 4x$ is

A. 270°
B. 180°
C. 90°
D. 45°

ANS: C DIF: Easy OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions KEY: period | sinusoidal function

4. The amplitude and period (in degrees) of $y = -2\sin 5x$ are

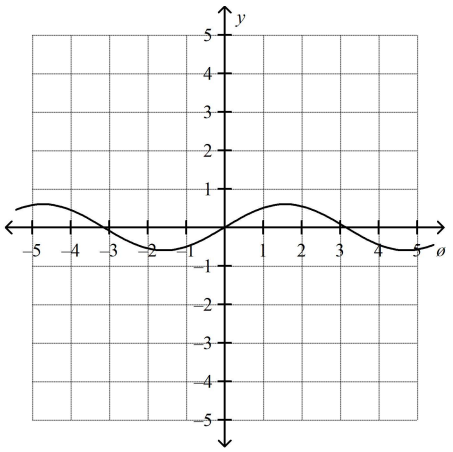
A. amplitude = $\frac{1}{5}$
period = 72°
B. amplitude = 2
period = 72°
C. amplitude = $-\frac{1}{5}$
period = 90°
D. amplitude = –2
period = 90°

ANS: B DIF: Average OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: amplitude | period | sinusoidal function

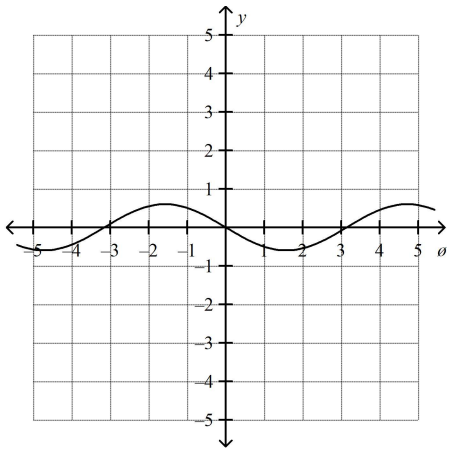
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5. Which graph represents the function $y = \frac{3}{5} \sin(-\theta)$, where θ is in radians?

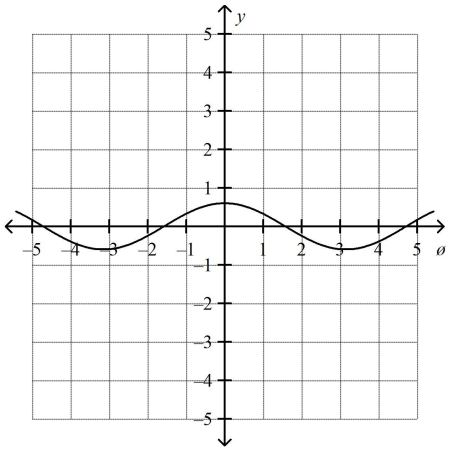
A.



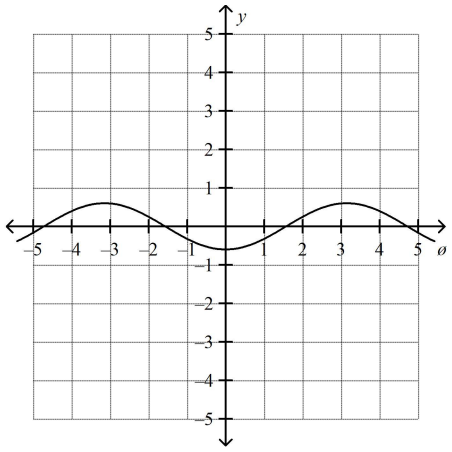
C.



B.



D.

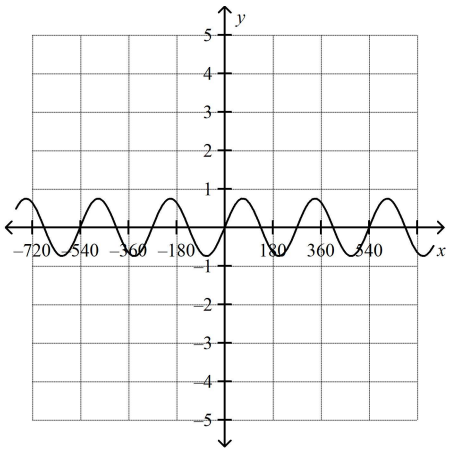


ANS: C DIF: Difficult OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: graph | amplitude | period | sinusoidal function

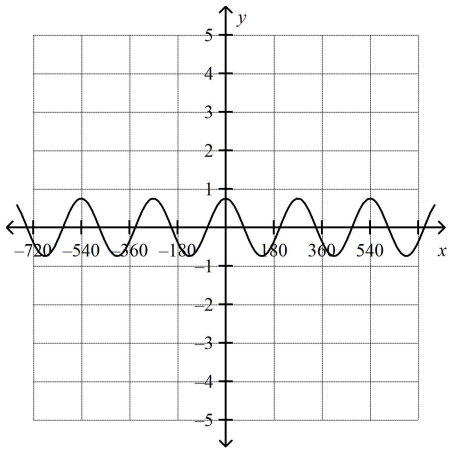
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6. Which graph represents the function $y = \frac{3}{4}\cos(-\frac{4}{3}x)$, where x is in degrees?

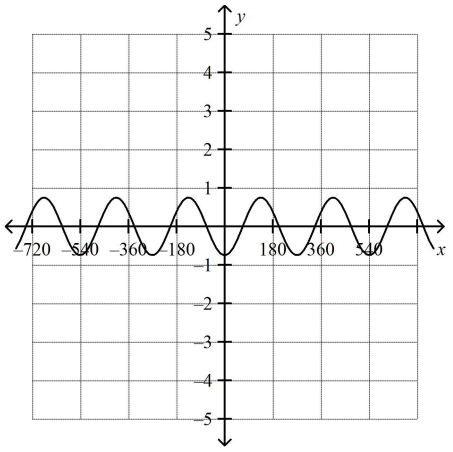
A.



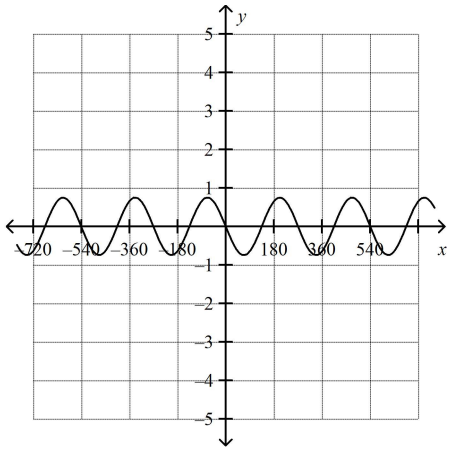
C.



B.



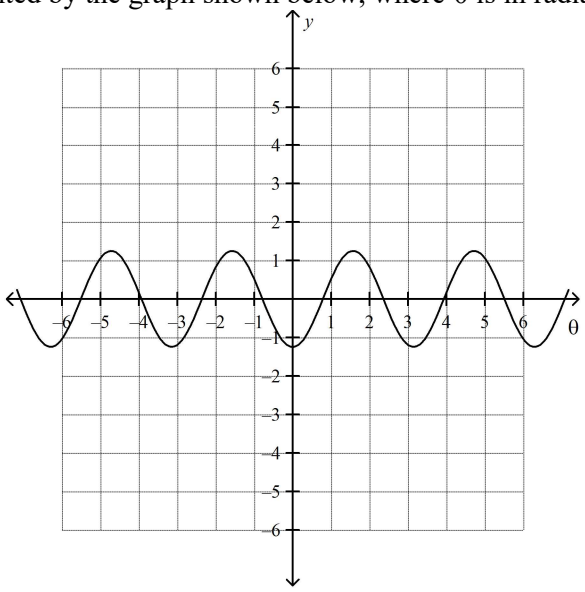
D.



ANS: C DIF: Difficult OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: graph | amplitude | period | sinusoidal function

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7. Which function is represented by the graph shown below, where θ is in radians?



A. $y = -\frac{5}{4}\sin(-2x)$

B. $y = -2\sin(-\frac{5}{4}x)$

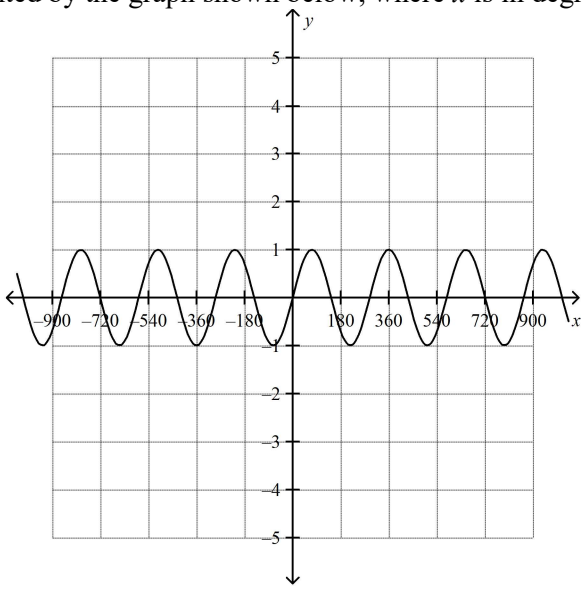
C. $y = -2\cos(-\frac{5}{4}x)$

D. $y = -\frac{5}{4}\cos(-2x)$

ANS: D DIF: Average OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: function | amplitude | period | sinusoidal function

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8. Which function is represented by the graph shown below, where x is in degrees?



A. $y = \sin(\frac{5}{4}x)$

B. $y = \frac{5}{4}\cos(x)$

C. $y = \cos(\frac{5}{4}x)$

D. $y = \frac{5}{4}\sin(x)$

ANS: A DIF: Average OBJ: Section 5.1 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: function | amplitude | period | sinusoidal function

9. The graph of $y = \sin x$ can be obtained by translating the graph of $y = \cos x$

A. $\frac{\pi}{4}$ units to the right

B. $\frac{\pi}{2}$ units to the right

C. $\frac{\pi}{3}$ units to the right

D. π units to the right

ANS: B DIF: Easy OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: translation | primary trigonometric function

10. What is the amplitude of the sinusoidal function $y = 7\sin\left(-8\left(x - \frac{\pi}{3}\right)\right) - 5$?

A. $\frac{\pi}{3}$

B. –8

C. –5

D. 7

ANS: D DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: amplitude | sinusoidal function

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11. What is the period of the sinusoidal function $y = -\cos\left(8\left(x - \frac{\pi}{2}\right)\right) - 2$?

A. $\frac{1}{8}\pi$

B. 4π

C. $\frac{1}{4}\pi$

D. $\frac{1}{2}\pi$

ANS: C DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions KEY: period | sinusoidal function

12. Determine the phase shift of the sinusoidal function $y = -6\sin\left(5\left(x - \frac{\pi}{3}\right)\right) + 2$.

A. $\frac{\pi}{3}$ units to the right

B. 3π units to the left

C. $\frac{\pi}{3}$ units to the left

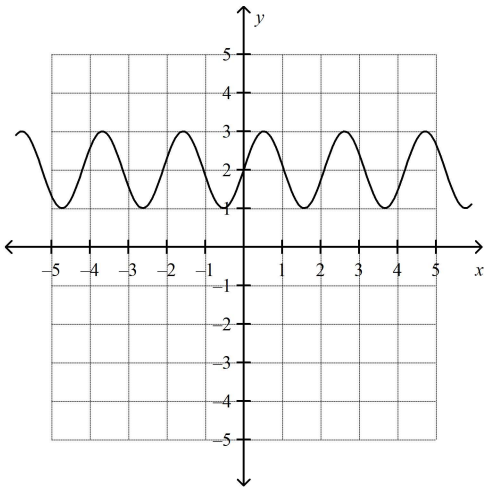
D. 3π units to the right

ANS: A DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: phase shift | sinusoidal function

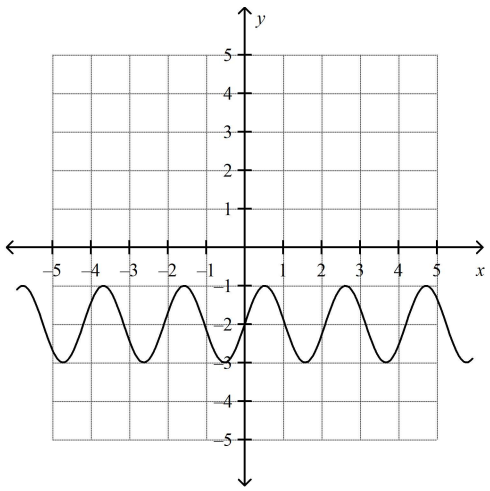
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13. Which graph represents the sinusoidal function $y = \cos\left(3\left(x + \frac{\pi}{6}\right)\right) - 2$?

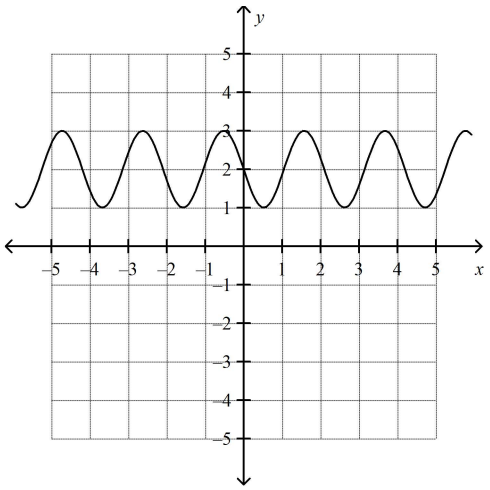
A.



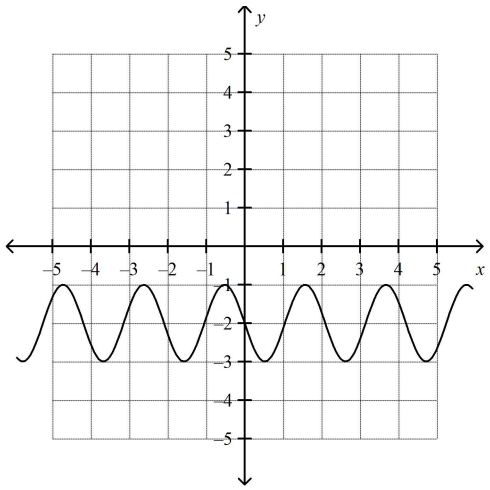
C.



B.



D.

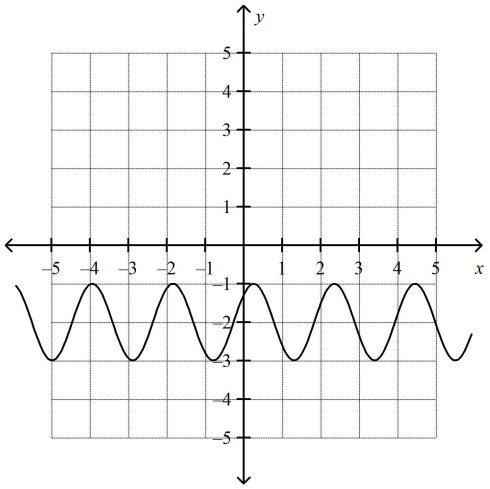


ANS: D DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: graph | sinusoidal function | transformation

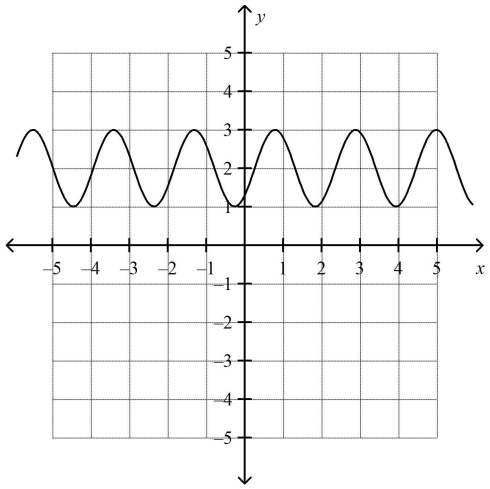
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14. Which graph represents the sinusoidal function $y = \sin\left(-3\left(x + \frac{\pi}{4}\right)\right) + 2$?

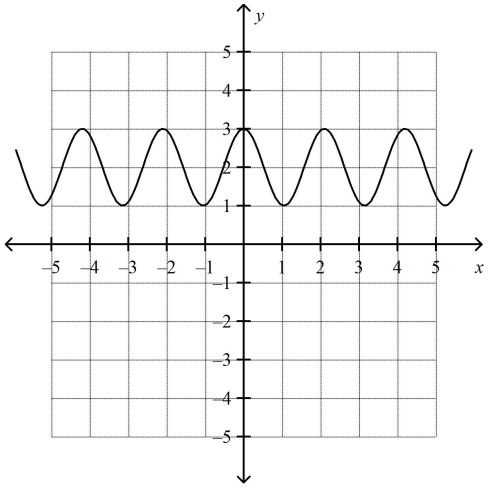
A.



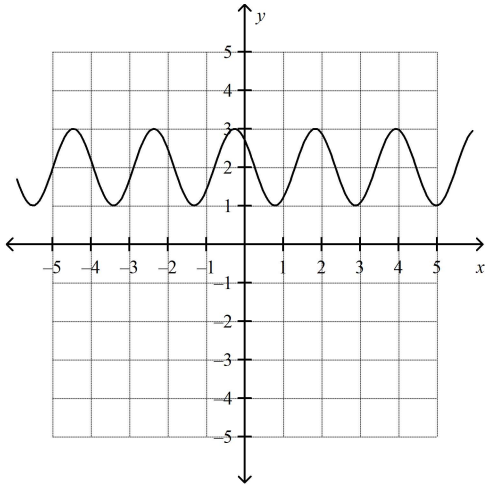
C.



B.



D.



ANS: C DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: graph | sinusoidal function | transformation

15. Give an equation for a transformed sine function with an amplitude of $\frac{5}{8}$, a period of $\frac{4}{5}$, a phase shift of $\frac{3}{2}$ rad to the right, and a vertical translation of 9 units down.

A. $y = \frac{5}{8}\sin\left[5/2(x + 3/2)\right] - 9$

C. $y = \frac{5}{8}\sin\left[5/2(x - 3/2)\right] - 9$

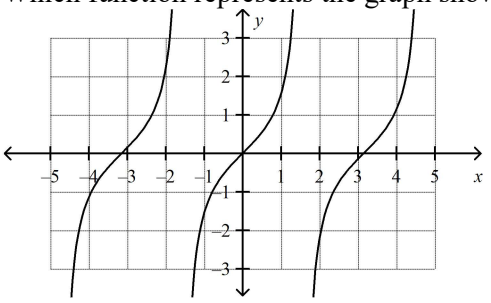
B. $y = \frac{5}{8}\sin\frac{5}{2}\pi(x + 3/2) - 9$

D. $y = \frac{5}{8}\sin\frac{5}{2}\pi(x - 3/2) - 9$

ANS: D DIF: Difficult OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: transformations | equation | properties | sinusoidal function

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16. Which function represents the graph shown, where x is in radians?



A. $y = \tan x$
B. $y = \cos x$
C. $y = -\tan x$
D. $y = \sin x$

ANS: A DIF: Easy OBJ: Section 5.3 NAT: T4
TOP: The Tangent Function KEY: graph | tangent function

17. Which of the following is not an asymptote of the function $f(\theta) = \tan\theta$?

A. $x = -\frac{7}{2}\pi$
B. $x = -\frac{9}{2}\pi$
C. $x = -\frac{5}{2}\pi$
D. $x = -\pi$

ANS: D DIF: Easy OBJ: Section 5.3 NAT: T4
TOP: The Tangent Function KEY: asymptote | tangent function

18. Which function has zeros only at $\theta = n\pi, n \in \text{I}$?

A. $y = \tan(\theta + \frac{2}{3}\pi)$
B. $y = \tan(\theta - \pi)$
C. $y = \tan(\theta - \frac{7}{6}\pi)$
D. $y = \tan(\theta + \frac{5}{4}\pi)$

ANS: B DIF: Difficult + OBJ: Section 5.3 NAT: T4
TOP: The Tangent Function KEY: zeros | transformation

19. Given the trigonometric function $y = \tan x$, which is the x-coordinate at which the function is undefined?

A. $\frac{9}{2}\pi$
B. $-\frac{7}{6}\pi$
C. $-\frac{1}{3}\pi$
D. $\frac{3}{4}\pi$

ANS: A DIF: Average OBJ: Section 5.3 NAT: T4
TOP: The Tangent Function KEY: undefined | tangent function

20. Given the trigonometric function $y = \tan x$, find the value of the y-coordinate of the point with x-coordinate –1200°.

A. $\sqrt{3}$
B. –1
C. 1
D. undefined

ANS: A DIF: Average OBJ: Section 5.3 NAT: T4
TOP: The Tangent Function KEY: coordinate | tangent function

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21. What are the solutions for $\sin^2 x - \frac{1}{2} = 0$ in the interval $0° \le x \le 360°$?

A. $x = 45°$ and $225°$ and $315°$ and $135°$
C. $x = 90°$ and $270°$ and $225°$
B. $x = 30°$ and $210°$ and $135°$
D. $x = 60°$ and $240°$ and $45°$

ANS: A DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: quadratic trigonometric equation

22. What are the solutions for $\cos^2 x - \frac{1}{2} = 0$ in the interval $0° \le x \le 360°$?

A. $x = 90°$ and $270°$ and $225°$
C. $x = 60°$ and $240°$ and $45°$
B. $x = 45°$ and $225°$ and $135°$ and $315°$
D. $x = 30°$ and $210°$ and $315°$

ANS: B DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: quadratic trigonometric equation

Use the following information to answer the questions.

The height, h, in metres, above the ground of a car as a Ferris wheel rotates can be modelled by the function $h(t) = 18\cos\left(\frac{\pi t}{80}\right) + 19$, where t is the time, in seconds.

23. What is the radius of the Ferris wheel?

A. 9 m
C. 19 m
B. 18 m
D. 36 m

ANS: B DIF: Easy OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: amplitude | sinusoidal function

24. How long does it take for the wheel to revolve once?

A. $\frac{\pi}{80}$ s
C. 160 s
B. 80 s
D. $\frac{80}{\pi}$ s

ANS: C DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: period | sinusoidal function

25. What is the minimum height of a car?

A. 19 m
C. 160 m
B. 9 m
D. 80 m

ANS: B DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: minimum | sinusoidal function

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26. What is the maximum height of a car?

A. 19 m
B. 80 m
C. 160 m
D. 31 m

ANS: D DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: maximum | sinusoidal function

Use the following information to answer the questions.

The height, *h*, in centimetres, of a piston moving up and down in an engine cylinder can be modelled by the function $h(t) = 14\sin(80\pi t) + 14$, where *t* is the time, in seconds.

27. What is the piston's minimum height?

A. 14 cm
B. –14 cm
C. 0 cm
D. 7 cm

ANS: C DIF: Easy OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: minimum | sinusoidal function

28. What is the period?

A. $\frac{7}{40}$ s
B. 8 s
C. $\frac{1}{40}$ s
D. $\frac{1}{14}$ s

ANS: C DIF: Easy OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: period | sinusoidal function

29. Solve $6\cos x + 2 = 0$ to three decimal places on the interval $x \in [0, 2\pi]$.

A. $x = 0.340, x = 5.943$
B. $x = 1.231, x = 5.052$
C. $x = 1.911, x = 1.231$
D. $x = 1.911, x = 4.373$

ANS: D DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: linear trigonometric equation

30. Solve $-5\cos x + 3 = 0$, to the nearest tenth of a degree, if necessary, on the interval $x \in [0°, 360°]$.

A. $x = 53.1°, x = 126.9°$
B. $x = 126.9°, x = 233.1°$
C. $x = 36.9°, x = 323.1°$
D. $x = 53.1°, x = 306.9°$

ANS: D DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: linear trigonometric equation

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COMPLETION

1. For the function $y = a\cos b(x - c) + d$, d affects the _____ of the graph.

ANS: vertical displacement

DIF: Easy OBJ: Section 5.1 | Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions | Transformations of Sinusoidal Functions
KEY: amplitude | period | phase shift | vertical displacement | sinusoidal function

2. The period of the basic sine function is _____ .

ANS: 2π

DIF: Easy OBJ: Section 5.1 | Section 5.3 NAT: T4
TOP: Graphing Sine and Cosine Functions | The Tangent Function
KEY: period | primary trigonometric function

3. The maximum value of the function $y = 3\cos\left(\frac{\pi}{6}x\right) - 9$ is _____ .

ANS: –6

DIF: Average OBJ: Section 5.2 | Section 5.3 NAT: T4
TOP: Graphing Sine and Cosine Functions | The Tangent Function
KEY: maximum | primary trigonometric function

4. The period of the function $y = -8\sin\left(\frac{\pi}{3}(x - 9)\right) - 3$ is _____ .

ANS: 6

DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions KEY: period | sinusoidal function

5. The phase shift of the function $y = -8\sin\left(\frac{\pi}{12}(x - 7)\right) + 2$ is _____ unit(s) to the _____.

ANS: 7, right

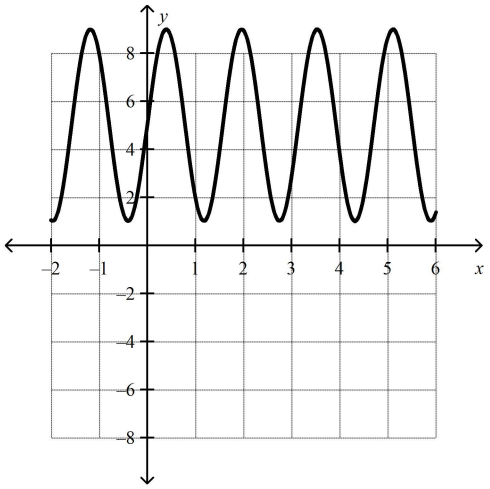
DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: phase shift | sinusoidal function

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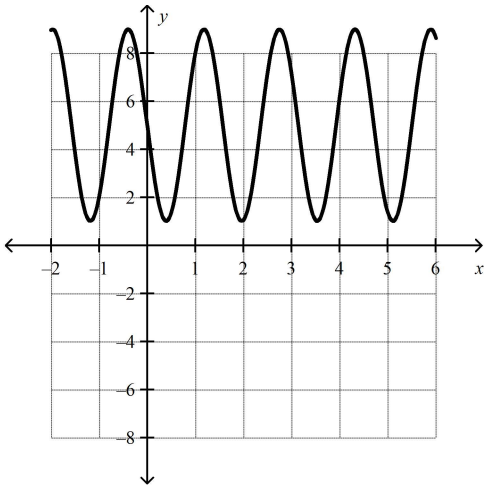
MATCHING

Match each equation with its graph.

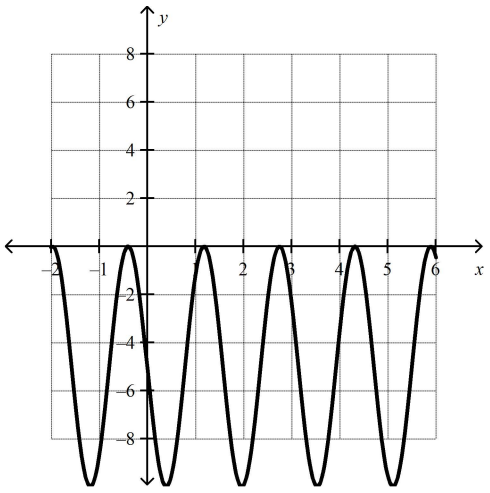
A.



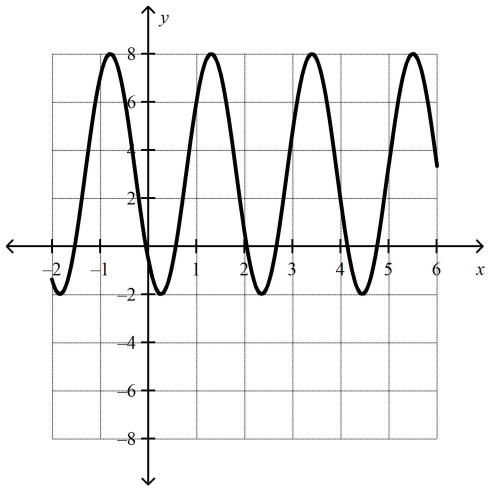
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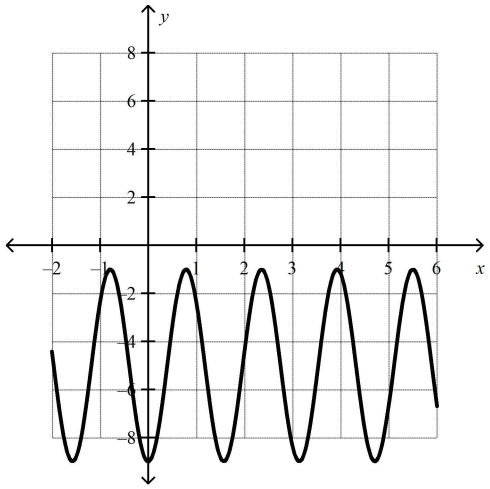
C.



D.



E.



1. $y = -4\sin\left[4\left(x + \frac{\pi}{4}\right)\right] + 5$

2. $y = -5\sin\left[4\left(x - \frac{\pi}{2}\right)\right] - 5$

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3. $y = 4\cos\left[4\left(x - \frac{\pi}{4}\right)\right] - 5$

4. $y = 5\cos\left[-3\left(x + \frac{\pi}{4}\right)\right] + 3$

1. ANS: A DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: transformations | sinusoidal function | graph | equation

2. ANS: C DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: transformations | sinusoidal function | graph | equation

3. ANS: E DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: transformations | sinusoidal function | graph | equation

4. ANS: D DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: transformations | sinusoidal function | graph | equation

SHORT ANSWER

1. Explain how you could graph the function $y = \cos x$ given a table of values containing ordered pairs for the function $y = \sin x$.

ANS:

You would need to subtract $\frac{\pi}{2}$ or 90° from each x-value for $y = \sin x$ and plot the points using the corresponding y-values. The zeros of the sine function would become the maximum or minimum values of the cosine function.

DIF: Easy OBJ: Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions KEY: phase shift

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2. When a ball is thrown, the horizontal distance, d, in metres, travelled can be modelled by the equation $d = \frac{v^2}{g}\sin 2\theta$, where $g = 9.8$ m/s^2; v is the velocity with which the ball is thrown, in metres per second; and θ is the angle of elevation, in radians. If a ball is thrown at 30 m/s, what angle of elevation is required for the ball to travel 30 m?

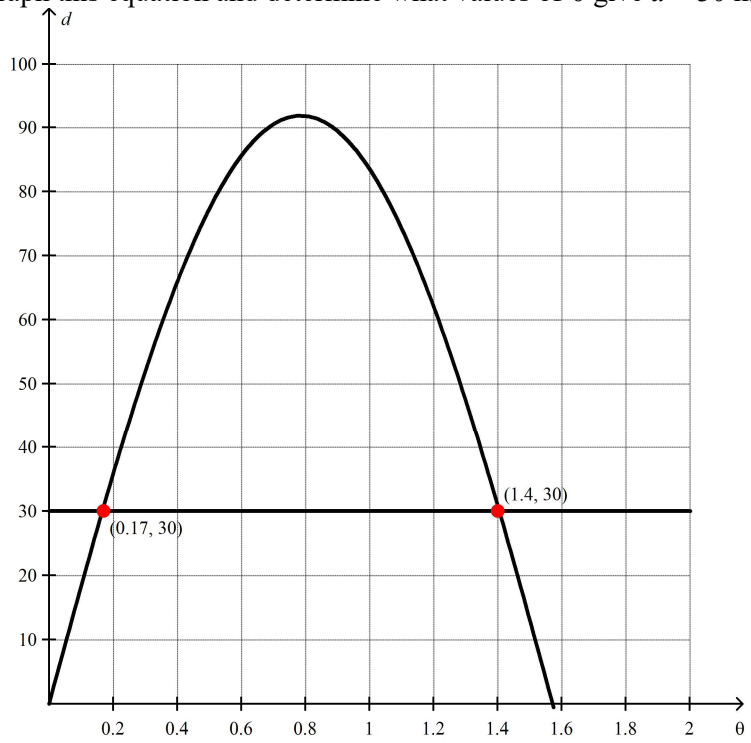
ANS:

$$d = \frac{v^2}{g}\sin 2\theta$$

$$= \frac{30^2}{9.8}\sin 2\theta$$

$$\approx 91.8\sin 2\theta$$

Graph this equation and determine what values of θ give $d = 30$ m.



From the graph, the ball would have to be thrown at an angle of approximately 0.17 radians or 1.4 radians.

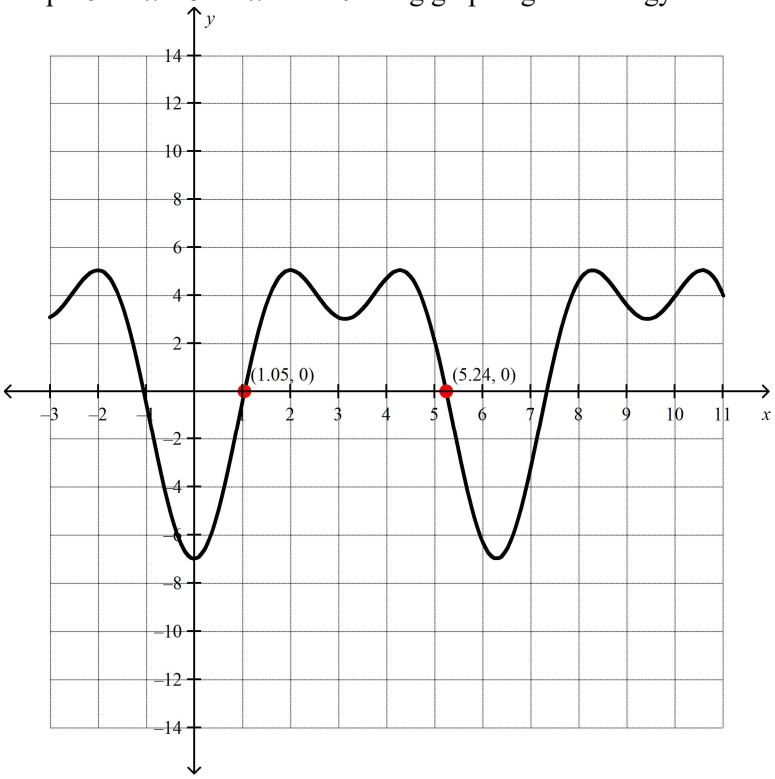
DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: trigonometric equation

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3. Solve $6\sin^2 x - 5\cos x - 2 = 0$ on the interval $x \in [0, 2\pi]$.

ANS:

Graph $6\sin^2 x - 5\cos x - 2 = 0$ using graphing technology.



The zeros occur at approximately 1.0 and 5.2 or exactly $\frac{\pi}{3}$ and $\frac{5\pi}{3}$.

DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: quadratic trigonometric equation

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4. A Ferris wheel of diameter 18.5 m rotates at a rate of 0.2 rad/s. If passengers board the lowest car at a height of 3 m above the ground, determine a sinusoidal function that models the height, h, in metres, of the car relative to the ground as a function of the time, t, in seconds.

ANS:
The amplitude is half the diameter of the Ferris wheel, or 9.25 m. The highest point is 18.5 + 3 or 21.5 m above the ground, so the vertical displacement is 21.5 − 9.25 or 12.25 m. The wheel rotates at a rate of 0.2 rad/s, which means it takes 5 s/rad. One full rotation of the wheel is equivalent to 2π radians. Therefore, it will take $2\pi\left(5\frac{\text{s}}{\text{rad}}\right) = 10\pi$ seconds for the Ferris wheel to do one full rotation.

The period is 10π.

$$\text{period} = \frac{2\pi}{|b|}$$

$$10\pi = \frac{2\pi}{|b|}$$

$$|b| = \frac{2\pi}{10\pi}$$

$$|b| = \frac{1}{5}$$

The value of b is thus $\frac{1}{5}$.

There is no phase shift if a cosine function with a negative value for a is used. Thus, the sinusoidal function that models the height of the car is

$$h = -9.25\cos\left(\frac{t}{5}\right) + 12.25$$

DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Graphing Sine and Cosine Functions
KEY: sinusoidal function | equation

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5. The water level at an ocean inlet has a depth, d, in metres, that varies with the time, t, in hours after midnight, according to the equation $d = 5\sin\left[2\pi\left(\frac{t-4}{12.4}\right)\right] + 6$. What is the water depth at 2:30 a.m., to the nearest hundredth of a metre?

ANS:
Substitute $t = 2.5$ into the equation, since 2:30 a.m. is 2.5 hours after midnight.

$$d = 5\sin\left[2\pi\frac{(t-4)}{12.4}\right] + 6$$

$$= 5\sin\left(2\pi\frac{(2.5-4)}{12.4}\right) + 6$$

$$\approx 2.56$$

The water depth is approximately 2.56 m.

DIF: Easy OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: sinusoidal function

6. Describe the transformations that, when applied to the graph of $y = \cos x$, result in the graph of $y = -2\cos\left[\frac{1}{8}\left(x - \frac{\pi}{3}\right)\right] + 1$.

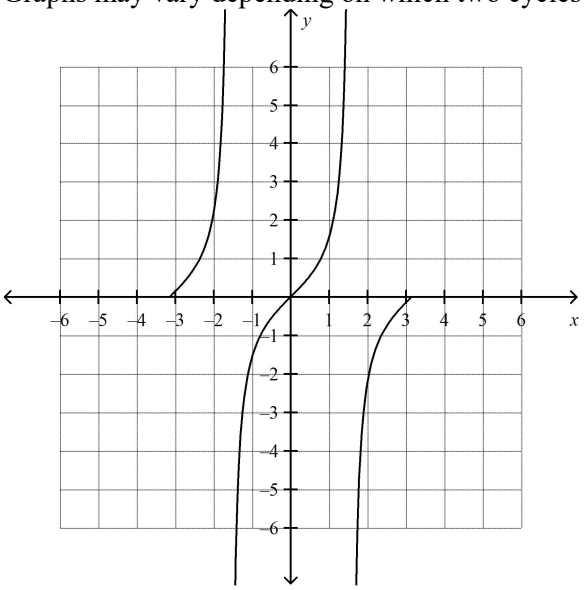
ANS:
a reflection in the x-axis, a vertical stretch by a factor of 2, a horizontal stretch by a factor of 8, a phase shift of $\frac{\pi}{3}$ to the right, and a vertical translation of 1 unit up

DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: transformations | sinusoidal function

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7. Sketch the graph of $y = \tan x$ for two cycles and state the domain, range, period, and equations of the asymptotes. x is measured in radians.

ANS:
Graphs may vary depending on which two cycles are graphed. Sample graph:



Domain: $\left\{x \mid x \neq \frac{\pi}{2} + n\pi, x \in \mathrm{R}, n \in \mathrm{I}\right\}$

Range: $y \in \mathrm{R}$

period: π

asymptotes: $x = \frac{\pi}{2} + n\pi, n \in \mathrm{I}$

DIF: Easy OBJ: Section 5.3 NAT: T4 TOP: The Tangent Function
KEY: tangent function | graph

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8. A pebble is embedded in the tread of a rotating bicycle wheel of diameter 60 cm. If the wheel rotates at 4 revolutions per second, determine a relationship between the height, h, in centimetres, of the pebble above the ground as a function of time, t, in seconds.

ANS:
Solutions may vary. Sample solution:

The amplitude is half the diameter, or 30 cm. The maximum height of the pebble is 60 cm, so the vertical displacement must be 30 cm. The wheel rotates at 4 revolutions per second, so the period is $\frac{1}{4}$ s. Thus, the value of b is $\frac{2\pi}{\frac{1}{4}}$ or 8π.

Thus, the relationship between the height of the pebble above the ground and time is
$h = 30\sin(8\pi t) + 30$

DIF: Easy OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: sinusoidal function | modelling

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9. A population, p, of bears varies according to $p(t) = 250 + 30\cos t$, where t is the time, in years, and angles are measured in radians.

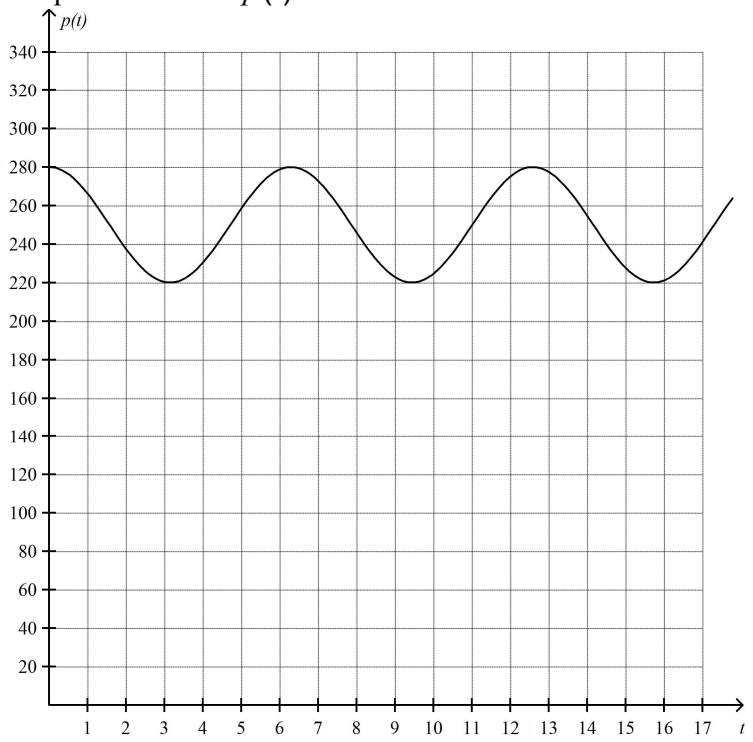
a) What are the maximum and minimum populations?

b) What is the first interval, in years and months, over which the population is increasing?

ANS:

a) From the function, the maximum and minimum populations are 250 + 30 or 280 bears and 250 − 30 or 220 bears.

b) Graph the function $p(t) = 250 + 30\cos t$.



The graph is first increasing over the interval [3.14,6.28] or [π, 2π], or from approximately 3 years $1\frac{2}{3}$ months to 6 years $3\frac{1}{3}$ months.

DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: sinusoidal function | maximum | minimum

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10. A girl jumps rope such that the height, h, in metres, of the middle of the rope can be approximated by the equation $h = 0.7\sin(72t + 9) + 0.75$, where t is the time, in seconds.
a) What is the amplitude of this function?
b) How many revolutions of the rope does the girl make in 1 min?

ANS:
a) 0.7 m

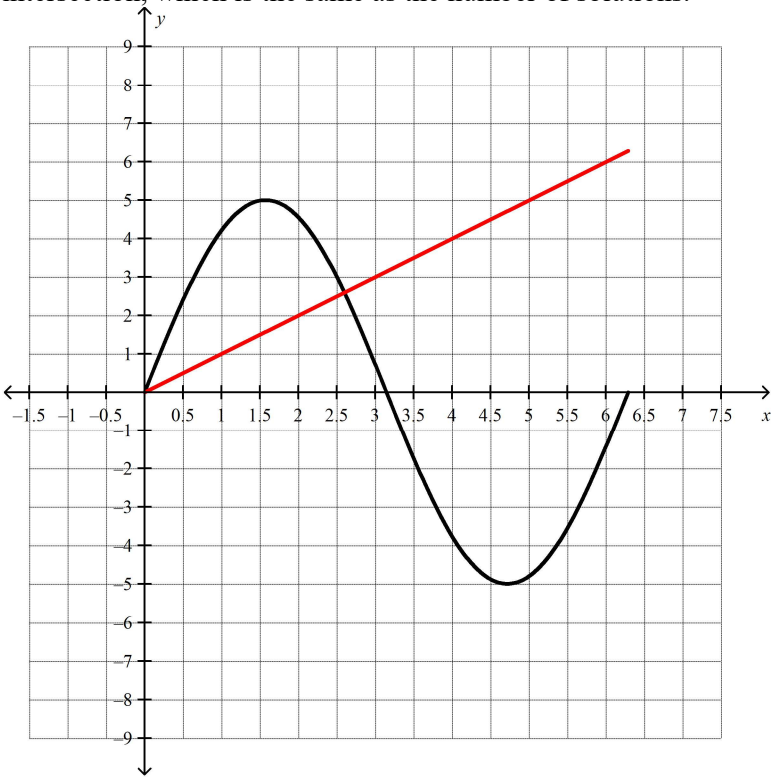
b) Since $b = 72$ and period $= \frac{2\pi}{b}$, then period $= \frac{2\pi}{72}$ or $\frac{\pi}{36}$ s. The number of revolutions of the rope is the reciprocal of the period, or $\frac{36}{\pi}$, or 11.46 rev/s. Multiply by 60 to get 688 revolutions/min.

DIF: Difficult OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: period | amplitude | sinusoidal function

PROBLEM

1. Determine the number of solutions for the trigonometric equation $5\sin x - x = 0$ on the interval $[0, 2\pi]$.

ANS:
Since $5\sin x = x$, a graph of the two functions $y = 5 \sin x$ and $y = x$ will show the number of points of intersection, which is the same as the number of solutions.



There are two solutions.

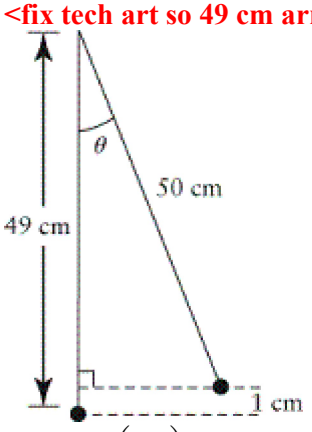
DIF: Difficult + OBJ: Section 5.3 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: linear trigonometric solution

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2. When a pendulum that is 0.5 m long swings back and forth, its angular displacement, θ, in radians, from rest position is given by $\theta = \frac{1}{4}\sin\left(\frac{\pi}{2}t\right)$, where t is the time, in seconds. At what time(s) during the first 4 s is the pendulum displaced 1 cm vertically above its rest position? (Assume the pendulum is at its rest position at $t = 0$.)

ANS:

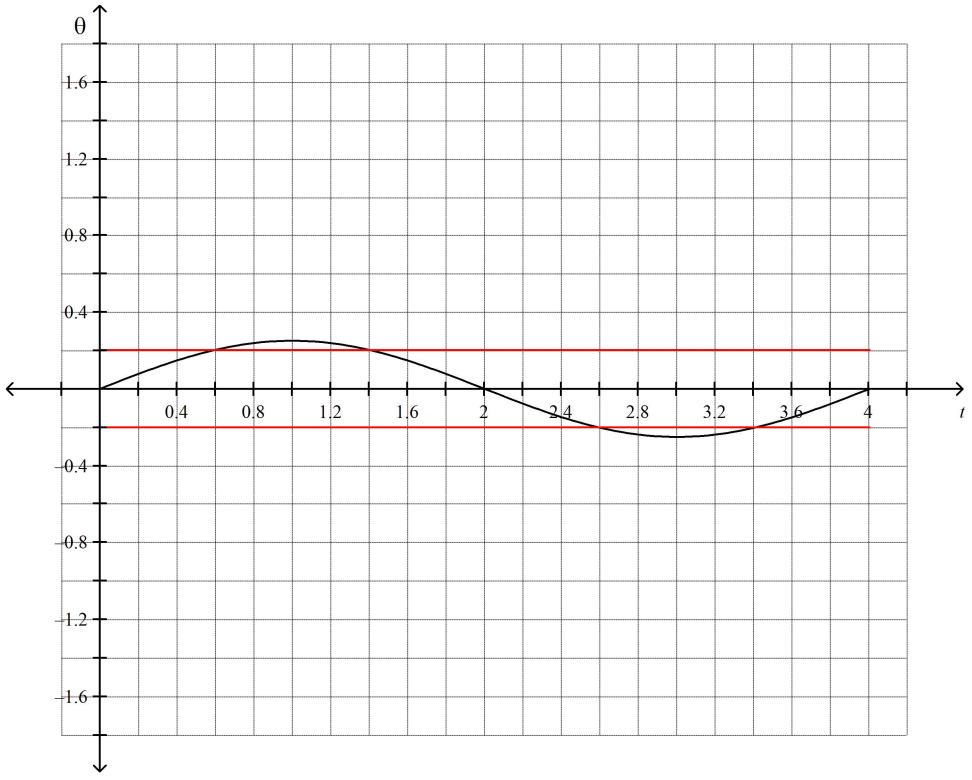
<fix tech art so 49 cm arrow goes only to bottom of right-angle triangle, not to black dot>



$\theta = \cos^{-1}\left(\frac{49}{50}\right)$

≈ 0.20033

Graph the function $\theta = \frac{1}{4}\sin\left(\frac{\pi}{2}t\right)$ and determine the points at which $\theta = \pm 0.20033$.



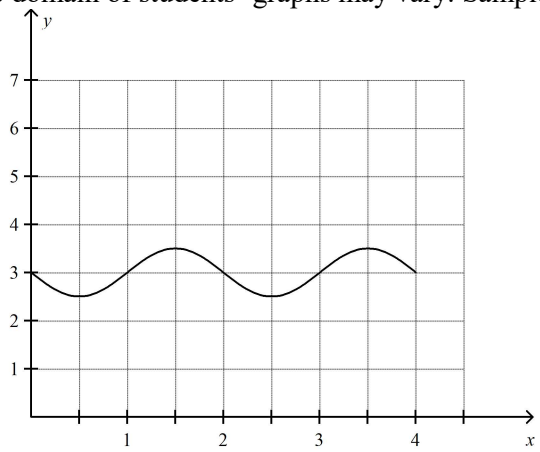
Therefore, during the first 4 s the pendulum is displaced 1 cm vertically at approximately 0.6 s and 1.4 s to one side and at approximately 2.6 s and 3.4 s to the other side.

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DIF: Difficult OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: linear trigonometric equation

3. Sketch the graph of $y = \frac{1}{2}\sin\left[\pi(x-1)\right] + 3$ for two cycles, where angles are in radians.

ANS:
The domain of students' graphs may vary. Sample graph:

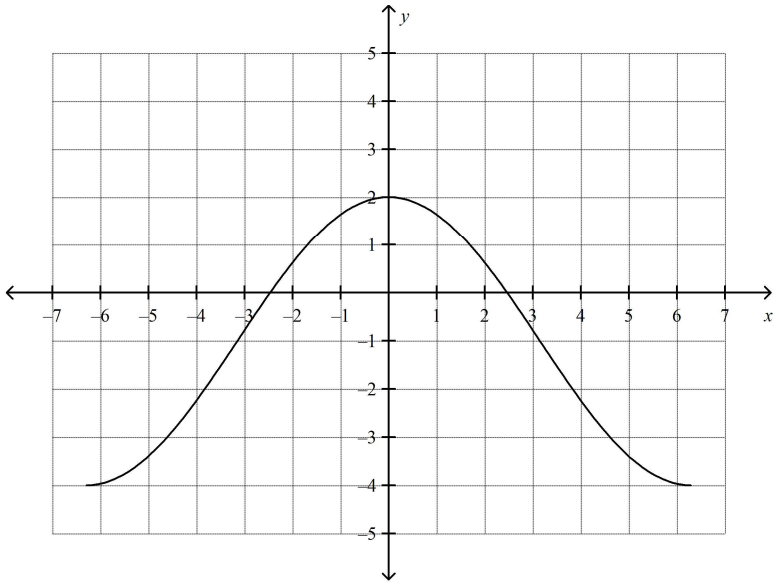


DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: sinusoidal function | graph | transformations

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4. Sketch the graph of $y = 3\cos\left(\frac{1}{2}x - 2\pi\right) - 1$ for $-2\pi \le x \le 2\pi$.

ANS:



DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: sinusoidal function | graph | transformations

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5. At a certain ocean bay, the maximum height of the water is 4 m above mean sea level at 8:00 a.m. The height is at a maximum again at 8:24 p.m. Assuming that the relationship between the height, *h*, in metres, and the time, *t*, in hours, is sinusoidal, determine the height of the water above mean sea level, to the nearest tenth of a metre, at 10:00 a.m.

ANS:
The period is 12 h 24 min, or 12.4 h.

$$12.4 = \frac{2\pi}{b}$$

$$b = \frac{\pi}{6.2}$$

An equation that models the depth of the water is $d = 4\cos\left(\frac{\pi}{6.2}t\right)$.

At 10:00 a.m., $t = 2$. Substitute $t = 2$ into the equation.

$$d = 4\cos\left(\frac{\pi}{6.2}t\right)$$

$$= 4\cos\left[\frac{\pi}{6.2}(2)\right]$$

$$\approx 2.1$$

The depth of the water at 10 a.m. is 2.1 m.

DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: equation | sinusoidal function | graph

6. A mass is suspended by a spring such that it hangs at rest 0.5 m above the ground. The mass is raised 40 cm and released at time $t = 0$ s, causing it to oscillate sinusoidally. If the mass returns to the high position every 1.2 s, determine the height of the mass above the ground at $t = 0.7$ s.

ANS:
The period is 1.2 s, so $b = \frac{2\pi}{1.2}$ or $\frac{5\pi}{3}$. The amplitude is 40 cm or 0.4 m. The vertical displacement is 0.5 m. Thus, the equation that models this situation is $h = 0.4\cos\left(\frac{5\pi}{3}t\right) + 0.5$.

Substitute $t = 0.7$ into the function to find *h*:

$$h = 0.4\cos\left(\frac{5\pi}{3}t\right) + 0.5$$

$$= 0.4\cos\left[\frac{5\pi}{3}(0.7)\right] + 0.5$$

$$\approx 0.15$$

The mass is about 15 cm above the ground at $t = 0.7$ s.

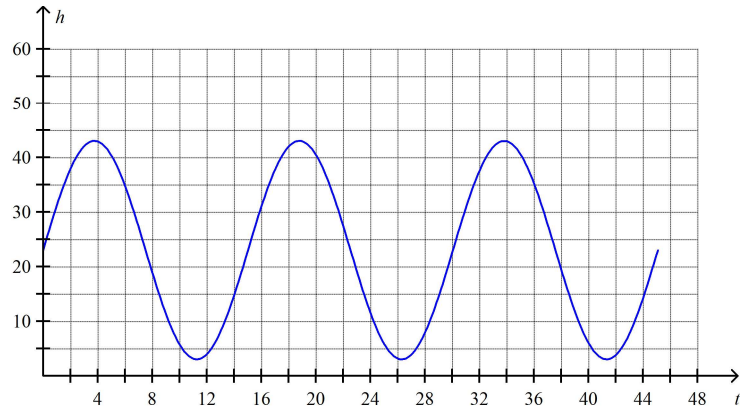
DIF: Difficult OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: sinusoidal function | graph | modelling

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7. A windmill has blades that are 20 m in length, and the centre of their circular motion is a point 23 m above the ground. The blades have a frequency of 4 revolutions per minute when in operation.
a) Use a sinusoidal function to model the height above the ground of the tip of one blade as a function of time.
b) Graph the function over three complete cycles.
c) How far above the ground is the tip of the blade after 10 s?

ANS:
Solutions may vary. Sample solution:
a) The amplitude is 20 m, and the vertical displacement is 23 m. The frequency of the blades is 4 revolutions per minute, so the period is 0.25 min or 15 s. Thus, $b = \frac{2\pi}{15}$, and the sinusoidal function is $h = 20 \sin\left(\frac{2\pi}{15} t\right) + 23.$

b)



c) Substitute $t = 10$ into the equation.

$$h = 20 \sin\left(\frac{2\pi}{15} t\right) + 23$$

$$= 20 \sin\left(\frac{2\pi}{15} (10)\right) + 23$$

$$\approx 5.7$$

The tip of the blade is approximately 5.7 m above the ground at $t = 10$ s.

DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: model | sinusoidal function | graph

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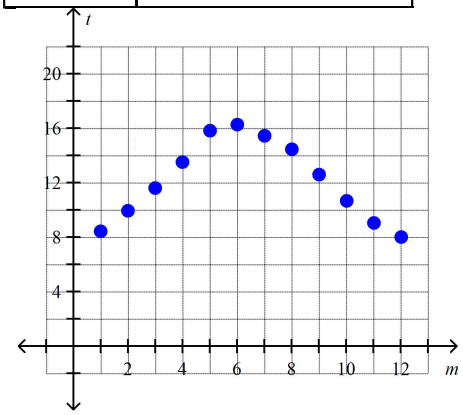
8. The table shows the hours of daylight measured on the first day of each month, over a 1-year period in a northern Ontario city.

Month	Hours of Daylight (h:min)
1	8:25
2	9:55
3	11:35
4	13:30
5	15:48
6	16:15
7	15:25
8	14:26
9	12:35
10	10:39
11	9:01
12	8:00

a) Graph the table data.
b) Use the graph and the table to develop a sinusoidal model to represent the information.
c) Graph the model on the same set of axes as the data. Comment on the fit.
d) Use your model to estimate the number of hours of daylight, to the nearest tenth of an hour, on January 15, and verify the solution using the graph.

ANS:
Answers may vary. Sample answer:
a) Convert h:min to hours.

Month	Hours of Daylight
1	8.42
2	9.92
3	11.58
4	13.5
5	15.8
6	16.25
7	15.42
8	14.43
9	12.58
10	10.65
11	9.02
12	8



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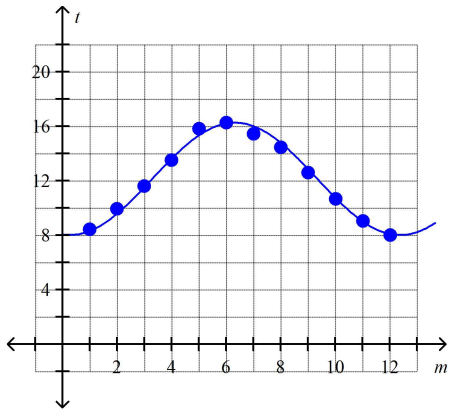
b) The amplitude is $\frac{16.25-8}{2}$, or $a = 4.125$. The vertical shift is 8 + 4.125, or $c = 12.125$.

The sine wave starts at 12.125, at approximately 3.29 months. So, $d = 3.29$.

The period is $\frac{360}{12}$, or $k = 30$.

An equation to represent the data is $t = 4.125 \sin [30(m-3.29)] + 12.125$.

c)



There seems to be a strong correlation between the graph and the data.

d) Substitute $m = 1.5$ (halfway between January 1 and February 1) into the equation:

$t = 4.125 \sin [30(m-3.29)] + 12.125$

$= 4.125 \sin [30(1.5-3.29)] + 12.125$

≈ 8.8

There are approximately 8.8 h, or 8 h 48 min, of daylight on January 15. The graph verifies this solution.

DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: model | sinusoidal function | graph

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9. The table shows the fraction of the Moon that can be seen at midnight from Simone's town. Day 1 represents January 1.

Day	1	2	3	4	5	6	10	14	19	21
Fraction Visible	0.25	0.17	0.12	0.06	0.02	0.00	0.10	0.56	0.98	1.00

Day	24	30	35	41	45	51	56	60	65	66
Fraction Visible	0.78	0.33	0.02	0.15	0.65	1.00	0.78	0.30	0.01	0.00

a) What is the period of the sine function that could be used to model the data?
b) What is the amplitude of the function?
c) What is the phase shift of the function?
d) What is the vertical shift?
e) Use your answers to parts a) to d) to write an equation for the function.
f) Use your function to determine the fraction of the moon visible to Simone on day
i) 100
ii) 150
iii) 200

ANS:
Answers may vary based on the technique used by the student (table data versus technology). Sample answer:
a) 30 days
b) 0.5
c) 14 days to the right
d) 0.5 up
e) $y = 0.5 \sin\ [12(x - 14)] + 0.5$
f) i) 0.13
ii) 0.40
iii) 0.98

DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions
KEY: model | sinusoidal function | amplitude | period | shifts

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10. The air quality index, I, in a large city can be modelled by the equation $I = 32 \sin [15(t-5)] + 28$, where t represents the time, in hours, after midnight. New legislation has been introduced that is expected to decrease the pollution levels in the city so that the index values will decrease to 90% of current values in 10 years.

a) What are the current minimum and maximum values of the index in the city?

b) At what time of the day is the air quality index at a maximum?

c) If an air quality alert is issued for times when the index is above 48, during what time period will an air quality alert be issued?

d) What factors in the equation for air quality index will be affected by the legislation? Explain how they will be affected.

e) What will the new air quality index equation be based on your answers in part d)?

f) How will these changes affect the times when an air quality alert will be issued?

ANS:

a) The minimum is –28 + 32 or –4, and the maximum is 32 + 28 or 60.

b) The maximum occurs when $\sin [15(t-5)] = 1$, which occurs when $15(t-5) = 90$. Thus,

$$15(t-5) = 90$$

$$t - 5 = 6$$

$$t = 11$$

The air quality index is at a maximum at 11 a.m.

c) Substitute $I = 48$ and solve for t.

$$I = 32 \sin [15(t-5)] + 28$$

$$48 = 32 \sin [15(t-5)] + 28$$

$$20 = 32 \sin[15(t-5)]$$

$$\frac{5}{8} = \sin[15(t-5)]$$

$$15(t-5) \approx 38.7 \text{ or } 15(t-5) \approx 141.3$$

$$t \approx 7.58 \text{ or } t \approx 14.42$$

$t = 7.58$ is equivalent to 7:35 a.m., and $t = 14.42$ is equivalent to 2:25 p.m. The air quality alert will be issued between 7:35 a.m. and 2:25 p.m.

d) Only the values of a and c will be affected by the legislation. The value of a will decrease from 32 to 28.8, and the value of c will decrease from 28 to 25.2.

e) $I = 28.8 \sin [15(t-5)] + 25.2$

f) Substitute $I = 48$ and solve for t.

$$I = 28.8 \sin [15(t-5)] + 25.2$$

$$48 = 28.8 \sin [15(t-5)] + 25.2$$

$$22.8 = 28.8 \sin[15(t-5)]$$

$$0.79 \approx \sin[15(t-5)]$$

$$15(t-5) \approx 52.2 \text{ or } 15(t-5) \approx 127.8$$

$$t \approx 8.48 \text{ or } t \approx 13.52$$

$t = 8.48$ is equivalent to 8:29 a.m., and $t = 13.52$ is equivalent to 1:31 p.m. The air quality alert will only need to be issued from 8:29 a.m. to 1:31 p.m.

DIF: Average OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions

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KEY: model | sinusoidal function | maximum | minimum | predict

11. George sells new cars at a local dealership. The table shows the number of cars sold by George each month for a year. Note that month 1 corresponds to January.

Month	Number of New Cars Sold
1	5
2	
3	20
4	30
5	37
6	39
7	36
8	30
9	19
10	10
11	4
12	2

a) Use the table to determine a function that can be used to model the information.
b) Over what domain and range is this model valid?
c) The value for the second month is missing. Use your model to determine how many cars George sold in that month.
d) Do you expect the data to continue to be sinusoidal? Explain.

ANS:
a) Answers may vary. Sample answer:
The amplitude is $\frac{39-2}{2}$, or $a = 18.5$. The vertical shift is 2 + 18.5, or $c = 20.5$.
The sine curve starts at 20.5, which is approximately month 3. So, $d = 3$.
The period is $\frac{360}{12}$, or $k = 30$.
George's new car sales can be modelled by the equation $y = 18.5 \sin [30(x-3)] + 20.5$.
b) The domain is the months from 1 to 12 (that is, January to December). The range is whole numbers from 2 to 39.
c) Answers may vary. Sample answer:
Substitute $x = 2$ into the equation:

$$y = 18.5 \sin [30(x-3)] + 20.5$$
$$= 18.5 \sin[30(2-3)] + 20.5$$
$$= 11.25$$

If the model holds, George sold 11 cars in February.
d) The data can be considered to be sinusoidal, as more cars would tend to be sold in the summer than in the winter, but economic factors could play a role in the nature of the function used to describe the sales.

DIF: Difficult + OBJ: Section 5.4 NAT: T4
TOP: Equations and Graphs of Trigonometric Functions KEY: model | sinusoidal function

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12. The flapping of a bird's wing can be modelled by the function $y = 5 \sin 7200t + 5$, where y represents the distance the tip of the wing travels, in centimetres, and t represents the time, in seconds.
a) Determine the period of the motion of the wing.
b) Determine the amplitude, the minimum value, and the maximum value.
c) What are the first times after $t = 0$ that the tip of the wing reaches the minimum and maximum values?
d) Determine the position of the wing tip at
i) $\frac{1}{60}$ s
ii) $\frac{1}{20}$ s
iii) $\frac{1}{10}$ s

ANS:
a) The value of b is 7200. Let P represent the period:

$$P = \frac{360}{7200}$$

$$= \frac{1}{20}$$

The period is $\frac{1}{20}$ s.
b) From the equation, the amplitude is 5 cm, the minimum is 0 cm, and the maximum is 10 cm.
c) The first minimum occurs when $\sin 7200t = -1$, or when $7200t = 270$.

$$7200t = 270$$

$$t = \frac{3}{80}$$

The first minimum occurs at $\frac{3}{80}$ s.
The first maximum occurs when $\sin 7200t = 1$, or when $7200t = 90$.

$$7200t = 90$$

$$t = \frac{1}{80}$$

The first maximum occurs at $\frac{1}{80}$ s.
d) Use a calculator to substitute the values of t into the equation.
i) approximately 9.33 cm
ii) 5 cm
iii) 5 cm

DIF: Average OBJ: Section 5.1 | Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions | Transformations of Sinusoidal Functions
KEY: model | sinusoidal function | amplitude | maximum | minimum

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13. Wilson places a measuring tape on a pillar of a dock to record the water level in his local coastal community. He finds that a high tide of 1.77 m occurs at 5:17 a.m., and a low tide of 0.21 m occurs at 11:38 a.m.

a) Estimate the period of the fluctuation of the water level.

b) Estimate the amplitude of the pattern.

c) Predict when the next two high tides will occur.

d) Predict when the next two low tides will occur.

ANS:

a) 11:38 – 5:17 = 6:21

So, 6 h 21 min represents half of a full cycle. Therefore, a full cycle would have a period of 12 h 42 min, or 12.7 h.

b) The amplitude is $\frac{1.77 - 0.21}{2}$, or 0.78 m.

c) The next high tide will occur 12 h 42 min after 5:17 a.m., or at 5:59 p.m. The next after this will be 12 h 42 min after 5:59 p.m., or at 6:41 a.m.

d) The next low tide will occur 12 h 42 min after 11:38 a.m., or at 12:20 a.m. The next after this will be 12 h 42 min after 12:20 a.m., or at 1:02 p.m.

DIF: Average OBJ: Section 5.1 | Section 5.2 NAT: T4
TOP: Graphing Sine and Cosine Functions | Transformations of Sinusoidal Functions
KEY: model | periodic function | predict

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14. Consider the function $y = \sin x - \cos x$.

a) Predict the y-intercept of the function.

b) Predict the x-intercepts from 0° to 360°.

c) Explain why your answers in part b) can be found by solving $\tan x = 1$.

d) Graph the function for the interval $0° \le x \le 360°$.

ANS:

a) The y-intercept is –1.

b) The x-intercepts are 45° and 225°.

c) When solving for the x-intercepts, you are solving the equation $\sin x = \cos x$.

$$\sin x = \cos x$$

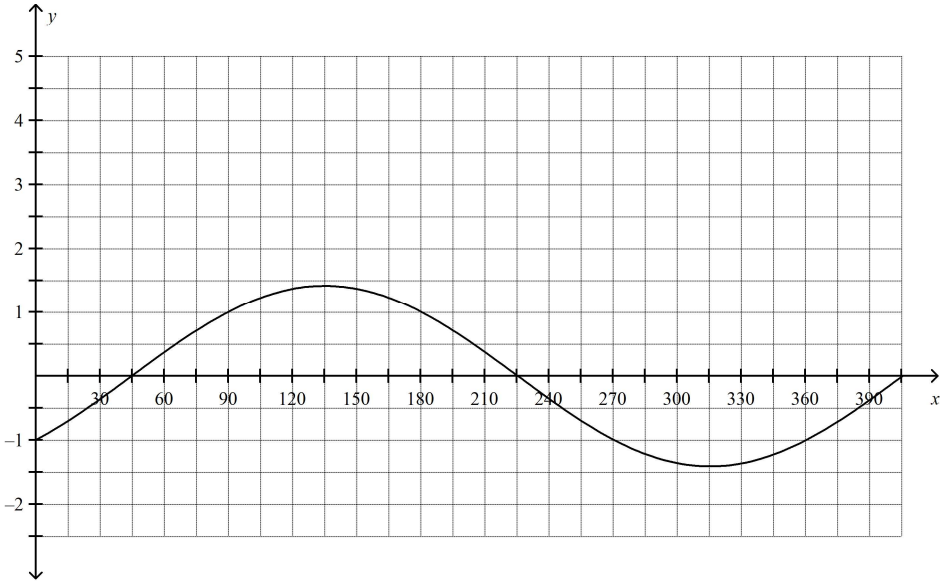
$$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = 45°$$

The second solution of 225° can be found by applying the CAST rule to a reference angle of 45° for the tangent ratio in the third quadrant.

d)



DIF: Difficult OBJ: Section 5.1 | Section 5.3 NAT: T4

TOP: Graphing Sine and Cosine Functions | The Tangent Function

KEY: sine function | cosine function | intercepts | graph

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15. Consider the graph of $y = 2\tan(x - 45°) + 3$.

a) Describe the transformations from the function $y = \tan x$.

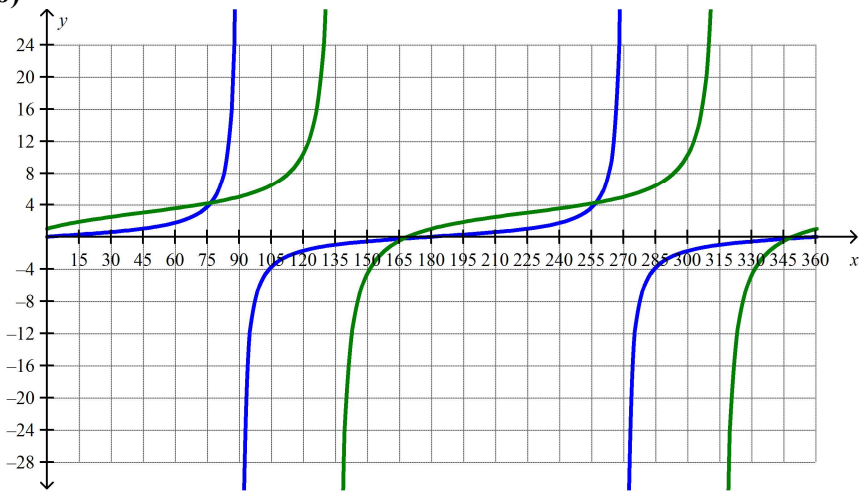
b) Graph the two functions on the same set of axes over the interval $0° \le x \le 360°$.

c) Given that $\tan x = \dfrac{\sin x}{\cos x}$, explain where the function $y = 2\tan(x - 45°) + 3$ has vertical asymptotes.

ANS:

a) The graph has an amplitude of 2, a phase shift of 45° to the right, and a vertical shift of 3 units up.

b)



c) The tangent function given has vertical asymptotes wherever the function $y = \cos(x - 45°)$ is equal to zero. This occurs at $x = 135°$ and $x = 315°$.

DIF: Difficult + OBJ: Section 5.3 NAT: T4 TOP: The Tangent Function
KEY: transformations | tangent function | asymptote

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16. The graph of $y = \cos x$ is transformed so that the amplitude becomes 2 and the x-intercepts coincide with the maximum values.

a) What is the equation of the transformed function?

b) What phase shift of the transformed function will produce a y-intercept of -1?

c) What is the equation of the function after the transformation in part b)?

d) Verify your solution to part c) by graphing.

ANS:

a) $y = 2\cos x - 2$

b) Let c represent the phase shift.

$y = 2\cos(x - c) - 2$

$-1 = 2\cos(0 - c) - 2$

$1 = 2\cos(-c)$

$\frac{1}{2} = \cos(-c)$

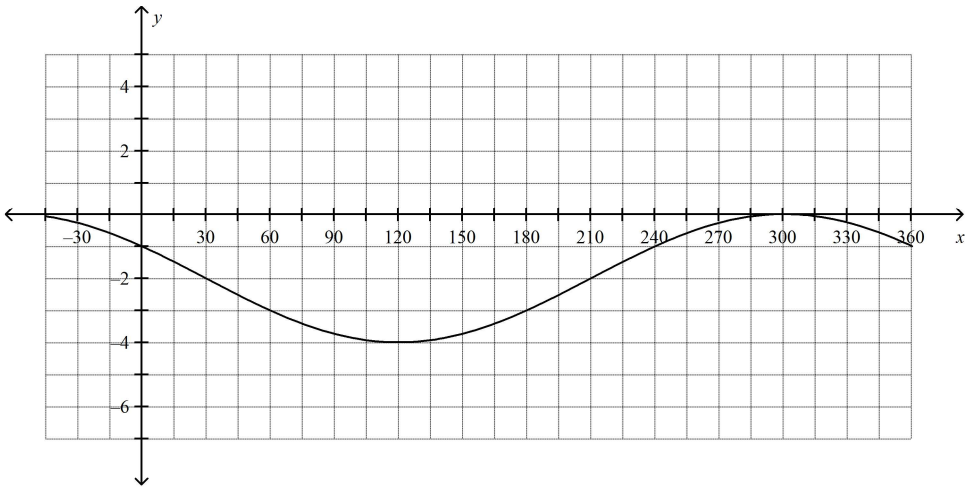
$-c = 60°$

$c = -60°$

The phase shift is 60° to the left.

c) $y = 2\cos(x + 60°) - 2$

d)



DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: graph | transformations | cosine function | equation

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17. A cosine function has half the period of the function $y = \cos x$. All other properties of the two functions are identical.

a) Determine the number of points of intersection if these two functions are graphed on the interval $0° \le x \le 360°$. Explain.

b) Determine the values of x of the first two points of intersection to the right of $x = 0°$.

c) If the graph of the two functions were extended to 720°, how many points of intersection would there be? Explain.

d) Use the concept of coterminal angles to determine the values of x where the points of intersection would occur.

e) Graph the two functions over the interval $0° \le x \le 720°$ to verify your answer for parts b), c), and d).

ANS:

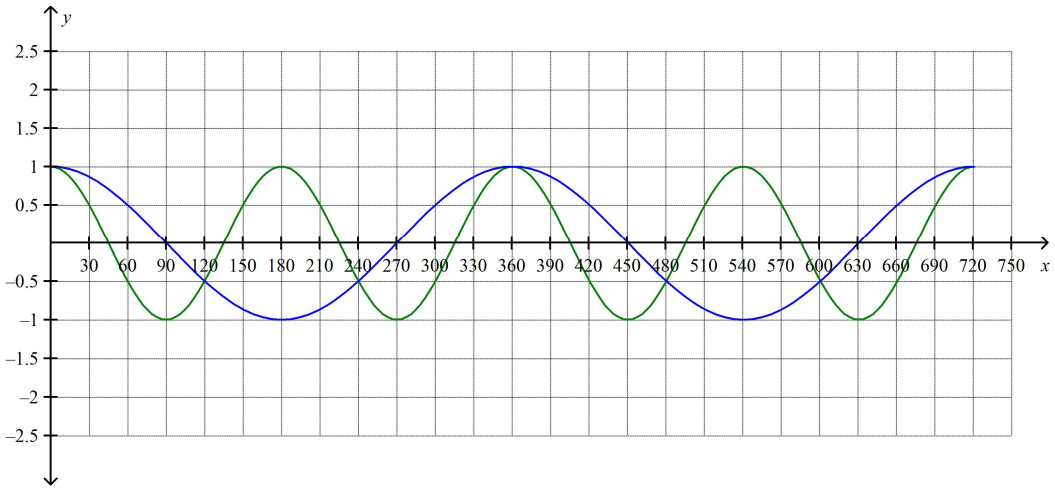
a) The two functions will have common maximum points at $x = 0°$ and $x = 360°$. Between these two points of intersection, there will be two others, as the cosine function with half of the period will repeat twice in the interval.

b) 120°, 240°

c) There would be points of intersections at 0°, 360°, and 720°, as well as the two in each interval between these points, for a total of seven points of intersection.

d) 0°, 120°, 240°, 360°, 480°, 600°, 720°

e)



DIF: Difficult OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: points of intersection | cosine function | coterminal angles

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18. A sinusoidal function has an amplitude of 2, a period of 180°, and a maximum at (0, 4).
a) Represent this function with an equation using a sine function.
b) Represent this function with an equation using a cosine function.
c) Explain how these two functions are related.

ANS:
a) The function $y = 2 \sin 2x + 2$ has an amplitude of 2, a period of 180°, and a maximum of 4, but the maximum occurs at $x = 45°$. For the maximum to occur at (0, 4), shift the function 45° to the left:
$y = 2 \sin [2(x + 45°)] + 2$
b) The function $y = 2 \cos 2x + 2$ has an amplitude of 2, a period of 180°, and a maximum of 4 at $x = 0$.
c) A phase shift of a sine function of 45° to the left will create a coincident cosine function, as long as the period of the sine function is 180°.

DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: equation | sine function | cosine function

19. Consider the function $f(x) = \frac{1}{2} \sin [3(x - 30°)] + 4$.

a) Determine the amplitude, the period, the phase shift, and the vertical shift of the function with respect to $y = \sin x$.
b) What are the minimum and maximum values of the function?
c) Determine the x-intercepts of the function in the interval $0° \le x \le 360°$.
d) Determine the y-intercept of the function.

ANS:
a) amplitude: $\frac{1}{2}$; period: $\frac{360°}{3}$ or 120°; phase shift: 30° to the right; vertical shift: 4 units up
b) minimum: $-\frac{1}{2} + 4$ or $\frac{7}{2}$, maximum: $\frac{1}{2} + 4$ or $\frac{9}{2}$
c) The function does not cross the x-axis, so there are no x-intercepts.
d) Substitute $x = 0$ in the function:

$$f(x) = \frac{1}{2} \sin[3(x - 30°)] + 4$$

$$f(0) = \frac{1}{2} \sin[3(0 - 30°)] + 4$$

$$= \frac{1}{2} \sin(-90°) + 4$$

$$= -\frac{1}{2} + 4$$

$$= \frac{7}{2}$$

The y-intercept is $\frac{7}{2}$.

DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: amplitude | period | shifts | transformations | sine function

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20. **a)** Describe the transformations that must be applied to the graph of $f(x) = \cos x$ to obtain the graph of $g(x) = -4 \cos 2x - 1$. Be sure to list the transformations in the proper order.
b) State the domain and range of $f(x)$ and $g(x)$.
c) Modify the equation for $g(x)$ to include a phase shift of 90° to the right. Name this equation $h(x)$.

ANS:
a) Apply the amplitude of 4 and reflect the function in the x-axis. Compress the function horizontally by a factor of 2. Shift the function 1 unit down.
b) $f(x)$: domain $\{x | x \in \mathrm{R}\}$, range $\{y | -1 \le y \le 1, y \in \mathrm{R}\}$
$g(x)$: domain $\{x | x \in \mathrm{R}\}$, range $\{y | -5 \le y \le 3, y \in \mathrm{R}\}$
c) $h(x) = -4 \cos [2(x - 90°)] - 1$

DIF: Average OBJ: Section 5.2 NAT: T4
TOP: Transformations of Sinusoidal Functions
KEY: transformations | domain | range | sine function

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